GeoQubit: variational quantum-circuit inversion of two-layer Wenner resistivity soundings

Team GeoQubit · Qiskit Fall Fest 2026 · https://github.com/HE1MER25/GeoQubit

Part 1. Project plan and team details

Member	Affiliation	Role
Praise Ayobami Ajulo-Omojesu	MSc Applied Geophysics, University of Ibadan	Project lead: geophysical theory, forward model, validation
John Wonderful Monday	BSc Physics, Obafemi Awolowo University	Circuit architecture, variational optimisation, simulation

Aim. We test whether a small variational quantum circuit, simulated in Qiskit and trained by a classical optimiser, can recover the reflection coefficient $k = (\rho_2 - \rho_1)/(\rho_2 + \rho_1)$ and top-layer thickness h of a two-layer earth from synthetic Wenner soundings, and compare it with bounded least squares on identical data. We do not claim a quantum advantage.

Objectives.

O1. Derive and validate the two-layer Wenner forward model.
O2. Generate soundings for six contrasts, noise-free and with 1–5 % noise.
O3. Build a 3-qubit circuit whose outputs decode to (k, h).
O4. Train it with COBYLA and SPSA from multiple random starts.
O5. Benchmark against least squares; diagnose initialisation and circuit size.

Phase	Output	Status
Theory and forward model	Derivation, Table 1	Complete
Circuit and optimisation	Figs. 1–2, Tables 2–3	Complete
Benchmark and noise study	Section 3.1–3.2	Complete
Diagnostics (after review)	Section 3.3–3.5	Complete

Risks. Local minima, barren plateaus and non-uniqueness are addressed by a small circuit, ten restarts, direct gradient measurements and a misfit landscape. Over-claiming is avoided by benchmarking against least squares on identical data. Deliverables: this report, the repository and a Colab notebook.

Part 2. Presentation, methodology, results and discussion

Key results. (1) The hybrid circuit recovered the true (k, h) in 60 of 60 noise-free runs. (2) Least squares needed 20 to 60 times fewer evaluations, so no quantum advantage is shown. (3) With noise, both methods give the same estimates, scattered along the equivalence valley. (4) The SPSA stall was a stationary point at $\boldsymbol{\theta} = \mathbf{0}$, not a barren plateau. (5) Six circuit angles suffice.

1 Problem and scope

A Wenner sounding injects current through two electrodes and measures potential across two others, all equally spaced by a. Larger a samples greater depth, and the curve of apparent resistivity ρ_a against a is inverted for the layering. We study the smallest case: two layers, known $\rho_1 = 100\,\Omega\,\mathrm{m}$, and unknowns k and h.

2 Methods

2.1 Forward model

The layer boundary and the free surface act as a pair of mirrors. The n-th image has weight k^n at depth $2nh$, and summing the images gives

$$\rho_a = \rho_1 \left[1 + 4 \sum_{n=1}^{\infty} k^n \left(\frac{1}{\sqrt{1 + (2nh/a)^2}} - \frac{1}{\sqrt{4 + (2nh/a)^2}} \right) \right]. \tag{1}$$

The geometric factor $K = 2\pi a$ is already absorbed, as the data are generated directly as ρ_a. The series is truncated when the tail bound $2\rho_1 |k|^{N+1}/(1 - |k|)$ falls below 10^{-8}. Data use $h = 8$ m and 12 log-spaced spacings from 1 to 50 m.

2.2 Validation

Equation (1) was compared with known limits and with the Hankel-transform form $\rho_a = \rho_1[1 + 4a \int_0^\infty K(\lambda)(J_0(\lambda a) - J_0(2\lambda a))\, d\lambda]$, $K = ke^{-2\lambda h}/(1 - ke^{-2\lambda h})$ (Table 1).

Test	Condition	Worst case ($\lvert k\rvert \le 0.9$)
Homogeneous	$k = 0$	$\rho_a/\rho_1 = 1$
Shallow limit	$a = 10^{-3}h$	7.9×10^{-10}
Deep limit	$a = 10^{4}h$	6.3×10^{-6}
Truncation	N versus $2N$	1.6×10^{-12}
Hankel integral	six contrasts	5.1×10^{-12}

Table 1. Validation of the forward model (relative errors).

2.3 Circuit, decoding and optimisation

Three qubits, three layers of R_Y, R_Z rotations and a CX ring give 18 angles $\boldsymbol{\theta}$. The expectation values $\langle Z_0 \rangle$, $\langle Z_1 \rangle$ are decoded as

$$\hat{k} = 0.95\, \langle Z_0 \rangle, \qquad \hat{h} = 1 + 29\, \frac{1 + \langle Z_1 \rangle}{2}\ \mathrm{m}, \tag{2}$$

and the misfit is the mean squared difference of $\log_{10}\rho_a$ between prediction and data. COBYLA used exact expectations (600 evaluations, 10 restarts); SPSA used 2000 emulated shots and 600 iterations. Shots are emulated by sampling the exact probabilities, and no hardware or Aer noise model was used. The baseline is SciPy bounded `least_squares` on (k, h) with the same misfit. The accompanying notebook re-runs every step with fixed seeds; its medians differ from the tables here in the last digits.

3 Results

3.1 Noise-free recovery

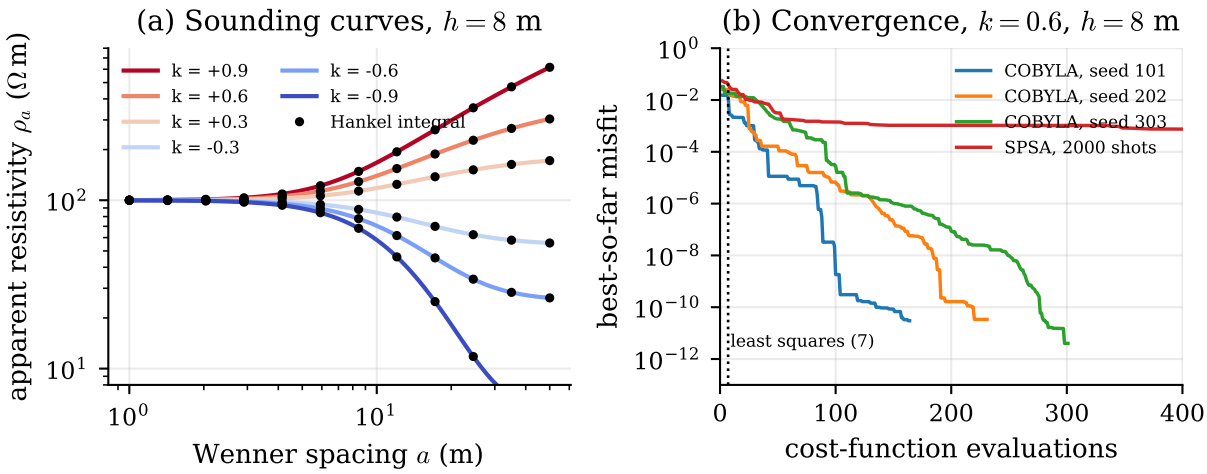


Figure 1. (a) Sounding curves for six contrasts (lines: image series; points: Hankel integral). (b) Best-so-far misfit for $k = 0.6$, $h = 8$ m: three independent COBYLA starts, one SPSA run (2000 shots) and least squares.

COBYLA reached 10^{-10} from every start in roughly 100 to 300 evaluations, SPSA levelled off near 10^{-3} to 10^{-4} because of shot noise, and least squares needed about seven. Both methods recovered the truth in all 60 runs (Table 2).

	Hybrid (COBYLA)			Least squares	
True k	Converged	Median evals	Median misfit	Converged	Median evals
$+0.9$	10/10	322	4×10^{-10}	10/10	10
$+0.6$	10/10	174	2×10^{-11}	10/10	7
$+0.3$	10/10	402	2×10^{-10}	10/10	7
-0.3	10/10	344	8×10^{-11}	10/10	7
-0.6	10/10	181	6×10^{-11}	10/10	6
-0.9	10/10	188	2×10^{-11}	10/10	8

Table 2. Noise-free recovery, $h = 8$ m; converged means $|\Delta k| < 0.01$, $|\Delta h| < 0.1$ m.

3.2 Equivalence and noise

Figure 2a shows a narrow valley through the true model: a thicker layer with larger k fits almost as well as a thinner one with smaller k. With 1, 3 and 5 % Gaussian noise (eight

realisations each), hybrid and least-squares estimates coincide and scatter along the valley (Table 3).

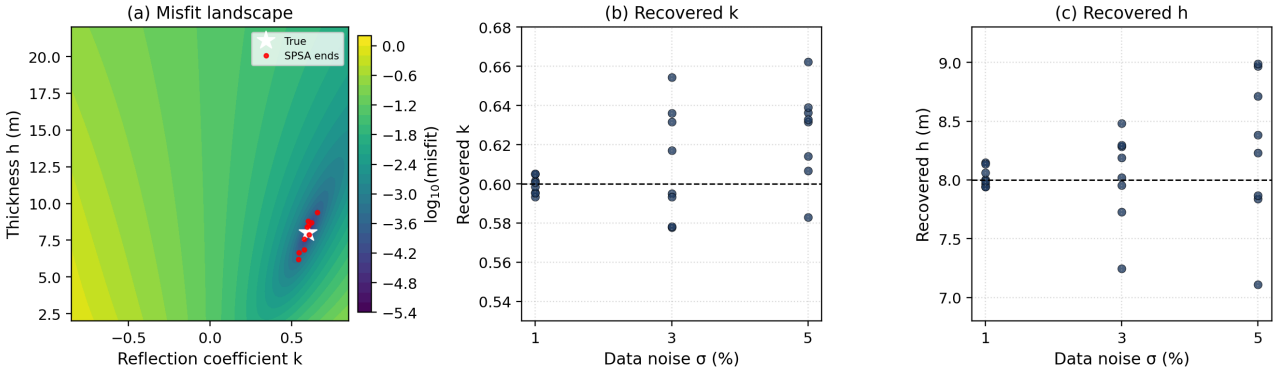


Figure 2. (a) Misfit landscape, $k = 0.6$, $h = 8$ m; red points are SPSA end points. (b, c) Recovered k and h for eight noisy datasets per noise level.

Noise	Hybrid k (0.6)	Hybrid h (8 m)	LS k	LS h (m)
1 %	0.599 ± 0.006	8.00 ± 0.11	0.599 ± 0.006	8.00 ± 0.11
3 %	0.600 ± 0.023	8.00 ± 0.48	0.600 ± 0.023	8.00 ± 0.48
5 %	0.619 ± 0.026	8.36 ± 0.42	0.619 ± 0.026	8.36 ± 0.42
SPSA	0.652 ± 0.016	8.52 ± 0.37	n/a	

Table 3. Mean $\pm$ s.d. over eight realisations (LS: least squares). SPSA: 2000 shots, ten seeds, no noise.

3.3 Initialisation and gradients

SPSA started near zero stalled at $\hat{k} \approx 0.95$, $\hat{h} \approx 30$ m. At $\boldsymbol{\theta} = \mathbf{0}$ every $\langle Z_q \rangle = 1$, which decodes to exactly these values, and the gradient is identically zero. This is a stationary point created by the decoding bounds, not a barren plateau (McClean et al., 2018), which is an exponential fall of gradient variance with qubit number. Over 150 random starts the median gradient norm was 0.003 near zero, 0.017 for $N(0, 0.3^2)$, 0.089 for $N(\pi/2, 0.5^2)$ and 0.095 for uniform angles; starting near $\pi/2$ removed the stall. The variance of $\partial\langle Z_0 \rangle / \partial\theta_1$ for 3-layer circuits fell from 0.126 at 2 qubits to 0.002 at 8, about half per added qubit. At three qubits there is no barren plateau, but larger circuits would develop one.

3.4 Circuit size

The choice of 18 angles was a standard hardware-efficient default. We repeated the noise-free $k = 0.6$ inversion with one to four layers (Table 4). All sizes converged in 10 of 10 restarts, so extra angles bring no accuracy gain.

Layers (angles)	1 (6)	2 (12)	3 (18)	4 (24)
Converged	10/10	10/10	10/10	10/10
Median evaluations to 10^{-10}	131	191	172	231

Table 4. Circuit-size ablation (COBYLA, at most 800 evaluations). Counts vary between runs; the pattern is the result.

3.5 Free top-layer resistivity

Releasing ρ_1 as a third unknown (decoded from $\langle Z_2 \rangle$ over 10–1000 Ω m on a log scale) makes the problem harder for the hybrid loop. With a cap of 1500 evaluations, 6 of 10 restarts recovered (ρ_1, k, h) within $|\Delta\rho_1| < 2\ \Omega$ m, $|\Delta k| < 0.01$ and $|\Delta h| < 0.1$ m; the rest ended on the equivalence valley. Least squares recovered all three exactly in seven evaluations.

4 Discussion

A three-qubit circuit recovers k and h for all six contrasts, so it is expressive enough for this inverse problem. It does so at 20 to 60 times the cost of least squares, which works directly on two numbers and uses local slope information, whereas COBYLA must search an 18-dimensional space without gradients. The ablation shows that the extra angles do not explain the accuracy and are not the main source of the cost.

With noisy data the two methods return the same estimates, because both reach the best attainable fit; the scatter is set by the equivalence of the data, not by the optimiser. Finite shots add a separate floor for SPSA. The study is limited to two unknowns (three in Section 3.5), one array and synthetic data on a simulator, so its conclusions apply to this toy problem only.

5 Further work and conclusion

The most useful next steps are warm-starting the circuit from a classical solution, extending to three-layer models, testing on published soundings with independent interpretations, and repeating the finite-shot runs on a noisy backend.

We validated a two-layer Wenner forward model, built a working hybrid quantum–classical inversion, and benchmarked it against least squares on identical data. The hybrid method recovers the true model at a much higher cost, so no quantum advantage is shown. The result is a reproducible baseline, with the initialisation failure diagnosed and the circuit size tested, for larger multi-layer studies.

Part 3. Declaration of assistance and resources

Resource	Type	How it was used
Claude (Anthropic), Claude app and web interface	AI assistant	Explained quantum-computing and resistivity concepts; drafted the method-of-images derivation and checks; wrote and ran the Python for the forward model, circuit, optimisers, baseline, experiments, diagnostics, notebook and figures; drafted, formatted and revised this report. The team supplied the problem, scope and plan, directed the revisions and reviewed the results.
Python, NumPy, SciPy, Matplotlib	Software	Numerics, COBYLA and least squares, quadrature for the Hankel check, plotting.
Qiskit	Software	Circuit construction and exact statevector expectation values; shots emulated by sampling. No Aer simulator or hardware.
LaTeX	Software	Typesetting this report.
Data	Synthetic	All data come from the validated forward model; no external or field data.
Mentorship	Human	Dr Sophie Engineer advised on project direction, use of AI tools and overall approach.

Verification. AI-produced derivations and code were checked with the tests in Table 1, including an independent numerical solution, and the diagnostics in Sections 3.3–3.5 were run after the review. The repository and notebook use fixed seeds. The team takes responsibility for this report.

References.

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